

E03 — Group Anagrams (Hash Table)

Experiment: 3 | LeetCode: #49 | CO: CO2, CO3 | BT Level: BT5

Problem Statement

Given an array of strings `strs`, group the anagrams together. You can return the answer in any order.

An **anagram** is a word formed by rearranging the letters of another word, using all original letters exactly once.

Example 1:

Input: `strs = ["eat", "tea", "tan", "ate", "nat", "bat"]`
Output: `[["bat"], ["nat", "tan"], ["ate", "eat", "tea"]]`

Example 2:

Input: `strs = [""]`
Output: `[[""]]`

Example 3:

Input: `strs = ["a"]`
Output: `[["a"]]`

Concept Review: Anagram Detection

Two strings are anagrams if and only if they have **the same character frequencies**.

Key insight: Sorting two anagrams always produces the same string:

- "eat" sorted → "aet"
- "tea" sorted → "aet"
- "ate" sorted → "aet"

Use the **sorted string as a hash key** to group anagrams.

Solution 1: Sorted Key (Primary)

```
from collections import defaultdict

def groupAnagrams(strs):
    """
    Group anagrams using sorted string as hash key.
    Time: O(n · k log k)    n = number of strings, k = max string length
    Space: O(n · k)         for the hash table
    """
```

```

"""
anagram_map = defaultdict(list)

for s in strs:
    key = tuple(sorted(s))    # Sorting takes O(k log k)
    anagram_map[key].append(s)

return list(anagram_map.values())

```

Solution 2: Character Count Key (Optional)

Instead of sorting ($O(k \log k)$), build a **frequency tuple** in $O(k)$:

```

def groupAnagrams_v2(strs):
    """
    Group anagrams using character frequency tuple as key.
    Time:  $O(n \cdot k)$      $n$  = number of strings,  $k$  = max string length
    Space:  $O(n \cdot k)$ 
    """
    anagram_map = defaultdict(list)

    for s in strs:
        # Build a 26-element count tuple (one per lowercase letter)
        count = [0] * 26
        for c in s:
            count[ord(c) - ord('a')] += 1
        key = tuple(count)    # Hashable key: (2, 0, 0, ..., 1, ...)
        anagram_map[key].append(s)

    return list(anagram_map.values())

```

Detailed Trace

Input: ["eat", "tea", "tan", "ate", "nat", "bat"]

Trace (Solution 1: sorted key)

String	sorted(s)	key (tuple)	anagram_map state
"eat"	"aet"	('a','e','t')	{('a','e','t'): ["eat"]}
"tea"	"aet"	('a','e','t')	{('a','e','t'): ["eat","tea"]}
"tan"	"ant"	('a','n','t')	{('a','e','t'): [...], ('a','n','t'): ["tan"]}
"ate"	"aet"	('a','e','t')	{('a','e','t'): ["eat","tea","ate"], ...}
"nat"	"ant"	('a','n','t')	{('a','n','t'): ["tan","nat"], ...}
"bat"	"abt"	('a','b','t')	{('a','b','t'): ["bat"], ...}

Result: ["eat", "tea", "ate"], ["tan", "nat"], ["bat"] ✓

Trace (Solution 2: frequency key)

String Frequency count (a-z positions)	Key
"eat" a = 1, e = 1, t = 1	(1,0,0,0,1,0,...,1,0,0)
"tea" a = 1, e = 1, t = 1	(1,0,0,0,1,0,...,1,0,0) ← same key
"bat" a = 1, b = 1, t = 1	(1,1,0,0,0,...,1,0,0)

Complexity Comparison

Solution	Time	Space	Notes
Sort key	$O(n \cdot k \log k)$	$O(n \cdot k)$	Simple, readable
Count key	$O(n \cdot k)$	$O(n \cdot k)$	Faster for large k

Python Counter Approach (for understanding)

```
from collections import Counter

def groupAnagrams_counter(strs):
    """Using Counter for readability (not hashable directly – need tuple)
    anagram_map = defaultdict(list)
    for s in strs:
        key = tuple(sorted(Counter(s).items())) # e.g., (('a',1),('e',
        anagram_map[key].append(s)
    return list(anagram_map.values())
```

Test Suite

```
def test_group_anagrams():
    test_cases = [
        ("eat", "tea", "tan", "ate", "nat", "bat"),
        (["eat", "tea", "ate"], ["tan", "nat"], ["bat"])),
        ([], [[]])),
        (["a"], [["a"]]),
        (["ab", "ba", "abc", "cab", "bca"], [
            ["ab", "ba"], ["abc", "cab", "bca"]
        ])
    ]

    for strs, expected in test_cases:
        result = groupAnagrams(strs[:])
        # Compare as sorted sets (order of groups doesn't matter)
        result_sorted = sorted([sorted(g) for g in result])
        expected_sorted = sorted([sorted(g) for g in expected])
        status = "PASS" if result_sorted == expected_sorted else "FAIL"
        print(f"{status}: {strs} → {result_sorted}")

test_group_anagrams()
```

Key Takeaways

- **Canonical form** (sorted string or frequency count) is the standard technique to detect anagrams.
 - Using a **hash table** (dict) with the canonical form as key achieves $O(1)$ grouping per string.
 - `defaultdict(list)` simplifies the grouping pattern — no need to check key existence.
 - For large alphabets or Unicode, the character count approach scales better than sorting.
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References

- LeetCode #49 — Group Anagrams
- Skiena, *The Algorithm Design Manual*, Ch. 4.2 (Dictionaries and Hashing)